higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

6 December 2024 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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SECTION A: FLOW MEASUREMENT

QUESTION 1

1.1 In the variable area flowmeter, the differential pressure across a variable aperture is used to adjust the area of the aperture.

1.1.1 Name TWO typical types of variable area meter. (2)

1.1.2 Float shapes depend on the type of fluid to be measured.

Name THREE types of float used in variable area flow meters. (3)

1.2 Electromagnetic flowmeters, sometimes known just as magnetic flowmeters, are limited to measuring the volume flow rate of electrically conductive fluids.

1.2.1 State Faraday's law of electromagnetic induction. (3)

1.2.2 Make a neat labelled sketch of magnetic flow meter. (7)

1.2.3 Write down a mathematical expression for the induced voltage of a magnetic flow meter. ALL the symbols used must be define. (5)

[20]

QUESTION 2

2.1 What is the function of an integrator? (2)

2.2 Can the magnetic flowmeter measure the flow of fluids like steam, air or gases? (2)

2.3 Make a graphical sketch showing how pressure drops across an orifice plate. (5)

2.4 Define the following pressure tapping's with reference to their positions from the orifice plate.

2.4.1 Radius taps

2.4.2 Vena contractor taps

2.4.3 Corner taps

(3 × 2) (6)

[15]

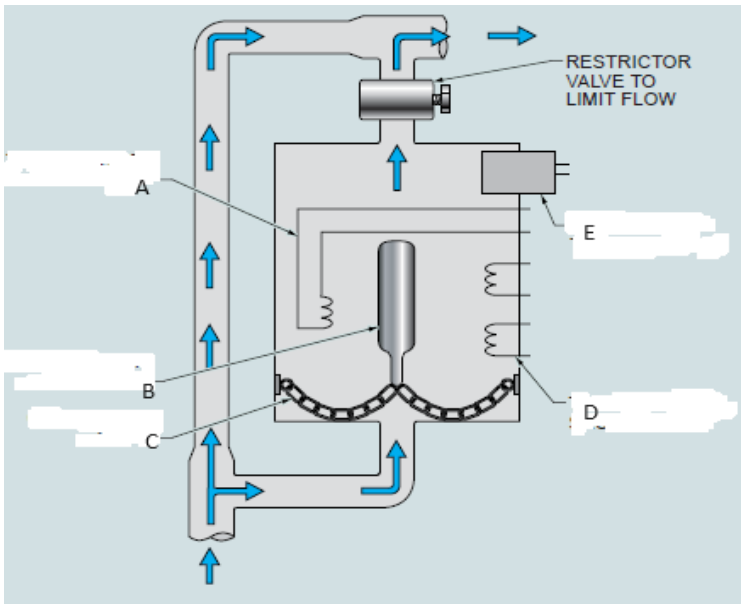
TOTAL SECTION A: 35

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SECTION B: DENSITY, HUMIDITY &VISCOSITY MEASUREMENT

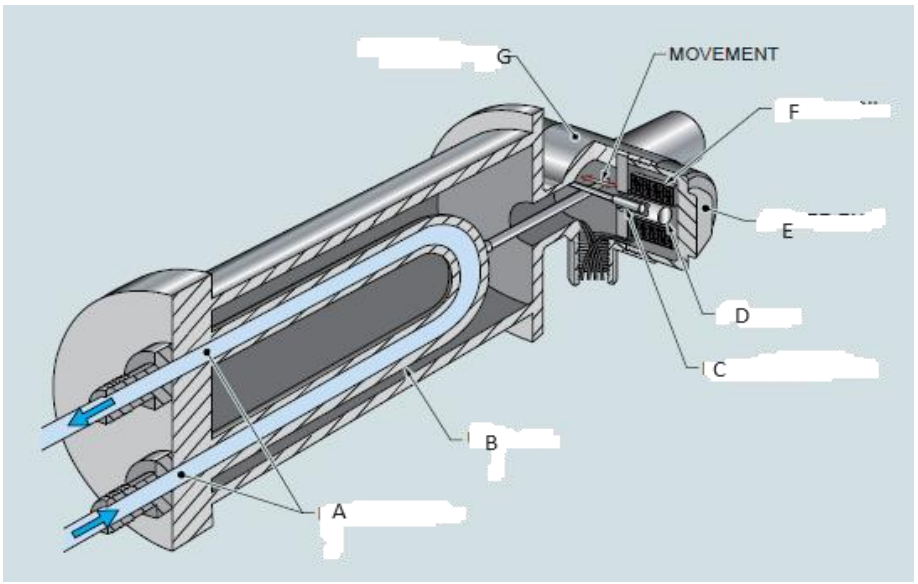
QUESTION 3

3.1 Continuous hydrometers are used for continuous measurement of density. Provide the labels for the indicated parts in the diagram below by writing only the name next to the letter (A–E) in the ANSWER BOOK.



(5 × 1) (5)

3.2 Vibrating U-tube density analysers determine density by measuring the natural vibration frequency of a tube filled with liquid. Name the labelled parts of the density analyser below by writing only the name next to the letter (A–G) in the ANSWER BOOK).

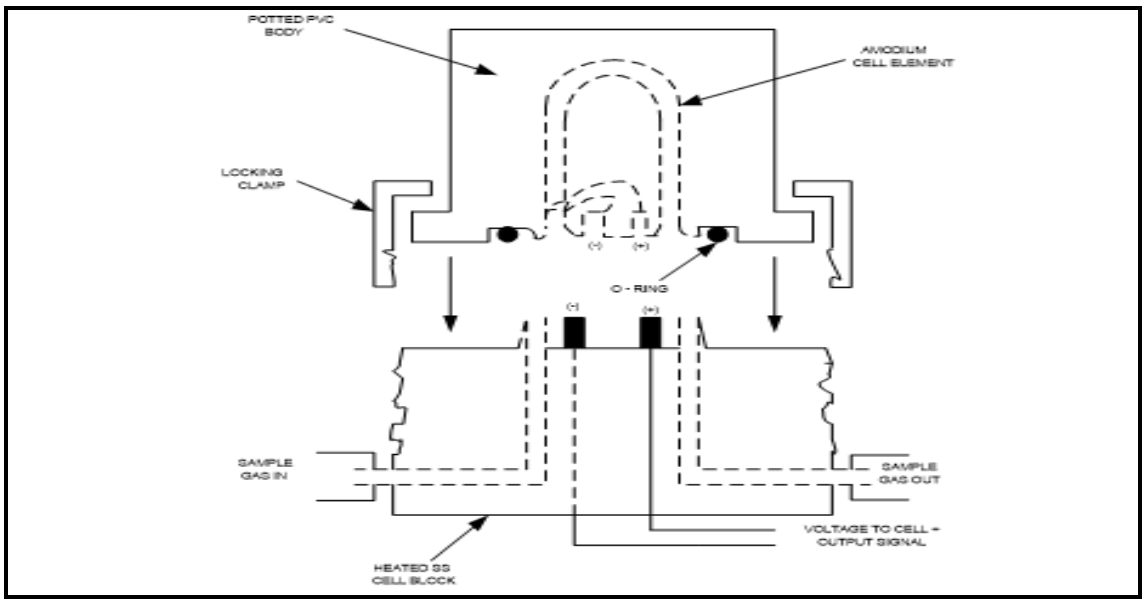


(7 × 1) (7)

3.3 Water absorbs electromagnetic radiation in the infrared region of the spectrum.

Make a neat schematic representation of the infra-red moisture detector. (6)

3.4 Study the diagram below and answer the questions that follow:



3.4.1 Name the hygrometer shown by the above diagram. (1)

3.4.2 Explain the operating principle of the hygrometer in QUESTION 3.4.1. (5)

[24]

QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 The emf from the pH electrode system is DC and the measuring instruments fall into two broad two classes of measuring circuits.

4.1.1 Briefly explain the TWO classes of measuring circuits. (2 × 2) (4)

4.1.2 List FIVE general requirements that must be met when using any of the measuring circuits for the pH electrode system. (5)

4.3 Metallic electrodes are subjected to poisoning by trace quantities of impurities in electrolytic solutions and by exposure to the atmosphere.

4.3.1 What is the main reason why precision measurements may be taken using glass electrode systems?

4.3.2 What can be done to make precision measurements using as glass electrode system?

(2 × 2) (4)

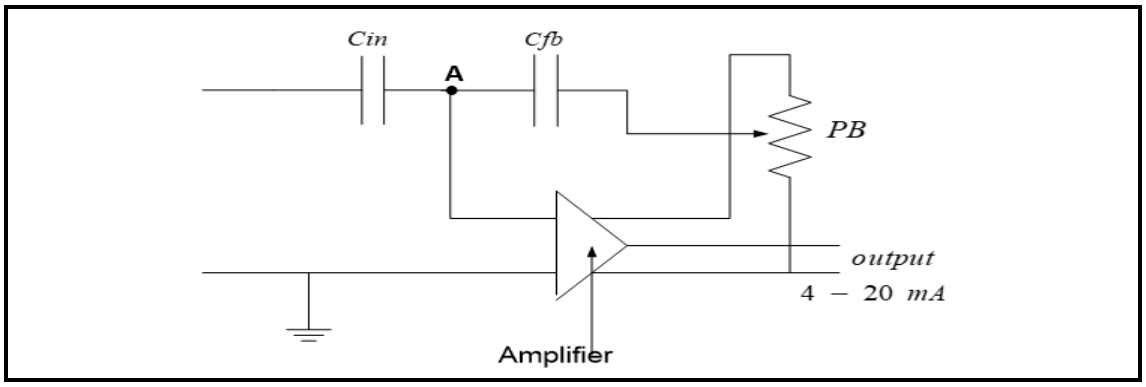
4.4 In order to maintain speed of response and accuracy, the lectrodes should be cleaned periodically.

Describe the process involved in the cleaning of pH electrodes. (6)

[19]

QUESTION 5: AUTOMATIC CONTROL

5.1 Below is a sketch of an electronic controller. Study the sketch and answer the questions the follow:



5.1.1 What type of control action is used by the above circuit? (1)

5.1.2 What happens when a potential appears at point? (3)

5.1.3 How does change the gain of this controller? (2)

5.2 Name the THREE control modes for electronic and electrically operated control systems. (3)

5.3 There are two forms of transmission lines in the process control field, namely electrical lines and pneumatic lines(pipelines), with electrical lines having advantages over pneumatic lines.

List FIVE broad assumptions to be considered for pneumatic transmission lines to be analogous to electrical lines. (5)

5.4 Define the following terms:

5.4.1 Position control

5.4.2 Direct acting

5.4.3 Process lag

5.4.4 Two position with inactive neutral

(4 × 2) (8)
[22]

TOTAL SECTION B: 65
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$W = 359 \cong 2\ CZ,Ed^2\sqrt{(h\rho)}$ $R_d = W/15 \cong 8\ \mu d$

$Q = 359 \cong 2\ CZ,Ed^2\sqrt{(h/\rho)}$ $R_d = Q\Delta/15 \cong 8\ \mu d$

$W = 0 \cong 01252\ CZ,Ed^2\sqrt{(h\rho)}$ $R_d = 3 \cong 54\ W/\mu d$

$Q = 0 \cong 01252\ CZ,Ed^2\sqrt{(h/\rho)}$ $R_d = 3 \cong 54\ Q\Delta/\mu d$

$Q_g = 2238\ CZEd^2\sqrt{(h/\rho)}$ $R_d = Q_g\Delta/98 \cong 6\ \mu d$

$m = (d/D)^2$ $E = 1/\sqrt{(1 - m^2)}$

$$N = \frac{W}{0 \cdot 01252 D^2\sqrt{(h\rho)}} = \frac{Q\sqrt{(\rho)}}{0 \cdot 01252\ D^2\sqrt{(h)}}$$

$mE = N/CZ,$ $CmE = N/Z,$ $mE = CmE/C$

$m = (d/D)^2$ $E = 1/\sqrt{(1 - m^2)}$

$$R_d = \frac{W}{15 \cdot 8\,\mu D\sqrt{(m)}} = \frac{Q\rho}{15 \cdot 8\ \mu D\sqrt{(m)}} = \frac{Q_g\rho}{98 \cdot 6\,\mu D\sqrt{(m)}}$$

$$N = \frac{W}{359 \cdot 2\ D^2\sqrt{(h)}} = \frac{Q\sqrt{(\rho)}}{359 \cdot 2\ D^2\sqrt{(h)}} = \frac{Q_g\sqrt{(\rho)}}{2238\ D^2\sqrt{(h)}}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$ $mE = N/CZ,$

$W = 1890\ Ud^2\sqrt{(\rho P)}$ *for critical flow/ vir kritieke vloei*

$CmE = N/Z,$ $mE = CmE/C$

$d = [W/1890U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *for critical flow/vir kritieke vloei*

$$R_d = \frac{3 \cdot 54W}{\mu D\sqrt{(m)}} = \frac{3 \cdot 54Q\rho}{\mu D\sqrt{(m)}}$$ $d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

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$W = 1{\cdot}252\,Ud^2\sqrt{(\rho P)}$ *for critical flow/ vir kritieke vloei*

$d = [W/1{\cdot}252\,U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *for critical flow/vir kritieke vloei*

1 kPa = 102 mmWD = 102 mmWG

1 lb/ft^3 = 16,0183 kg/m^3

Atmospheric pressure/Atmosferiese druk = 101,325 kPa

Gravitation acceleration/Gravitasieversnelling = 9,81 m/s^2

For/ Vir D + D/2 tappings and flange tappings/tappe en flenstappe

$$\frac{h}{Pa} \times 27{,}2 = \frac{kPa}{kPa} \times 27{,}2$$

$$Q = \frac{8}{15} \quad Tan\,\frac{\theta}{2}\sqrt{2g.H^5} \qquad\qquad Q = \frac{2}{3} \quad B\sqrt{2g.H^3}$$

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